FAST — NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES
FAST School of Computing Karachi Campus Department of Artificial Intelligence & Data Science
AL-2001 Programming for Artificial Intelligence (Laboratory) · Fall 2026

PROJECT PROPOSAL

Semester Lab Project Weightage 1 mark

Group Information

	Name	Roll Number	Section	Role in the project
Leader	Syeda Haram Waqar	25K-0116	BAI-3A	AI project Developer
Member 2	None			
Member 3	None			

Project Title	GlucoGuide AI – Diabetes Awareness and Risk Education System
Specialisation Track	Track A — Text / NLP
GitHub Repository URL	
Date of Submission	13th week

1. Project Title and Summary

GlucoGuide AI is a Python-based system designed to spread awareness about diabetes. It answers users' diabetes-related questions through a text-based health assistant. It also analyzes user-provided health information to explain possible risk factors for educational purposes. The system uses charts and visualizations to present health information clearly. It helps students and individuals understand diabetes and its risk factors, but it does not provide a medical diagnosis.

2. Problem Statement and Motivation

Many people lack clear and reliable information about diabetes, its risk factors, and prevention. Finding useful information online can be time-consuming because it is spread across different websites and often uses complicated medical terms. Students and individuals may struggle to understand diabetes-related information and interpret health data. Currently,

people rely on internet searches, articles, or informal advice to learn about diabetes, which can sometimes cause confusion or misunderstandings.

GlucoGuide AI aims to address this problem by providing a text-based assistant that answers diabetes-related questions, explains possible risk factors using user-provided information, and presents health information through simple charts. The system will provide educational guidance rather than medical diagnosis. Solving this problem is worthwhile because it can make diabetes awareness more accessible and encourage people to make informed decisions about seeking professional medical advice.

3. Proposed Solution

GlucoGuide AI will provide a simple and interactive way for users to learn about diabetes. Users will be able to enter diabetes-related questions or provide basic health information, such as age and other relevant risk factors. The system will process their input and provide easy-to-understand educational responses. The health assistant will answer common questions about diabetes, while the risk assessment component will explain possible risk factors based on the information provided. The visualization component will display relevant health data through charts, helping users understand patterns and information more easily. The system will be developed in Python using a Jupyter Notebook environment. It will focus on health education and awareness rather than providing medical diagnoses or replacing professional medical advice.

The main components of the project are:

1. **User Input Module:** Collects users' questions and basic health information.
2. **NLP Health Assistant:** Processes text-based questions and provides diabetes-related educational answers.
3. **Risk Assessment Module:** Examines the provided information and explains relevant risk factors.
4. **Data Visualization Module:** Creates charts to present health information clearly.
5. **Output Module:** Displays responses, risk-factor explanations, and visualizations in an understandable format.

3.1 Planned Class Design

1. HealthAssistant

- **Attributes:** `knowledge_base`, `user_question`

- **Methods:** `answer_question()`, `search_information()`
- **Purpose:** Answers users' diabetes-related questions with educational information.

2. RiskAssessment

- **Attributes:** `age`, `health_information`, `risk_factors`
- **Methods:** `collect_information()`, `assess_risk_factors()`, `generate_explanation()`
- **Purpose:** Examines user-provided information and explains relevant diabetes risk factors for educational purposes.

3. DataVisualizer

- **Attributes:** `health_data`, `chart_type`
- **Methods:** `load_data()`, `create_chart()`, `display_visualization()`
- **Purpose:** Presents health-related data through charts and graphs.

4. GlucoGuideAI

- **Attributes:** `assistant`, `risk_assessor`, `visualizer`
- **Methods:** `run_system()`, `display_results()`
- **Purpose:** Coordinates the main components and displays the results to the user.

Relationships: The `GlucoGuideAI` class coordinates the `HealthAssistant`, `RiskAssessment`, and `DataVisualizer` classes. Each class performs its own task, and the main class brings their functionality together into one system.

4. Scope

In scope — the features you commit to delivering. List the features that must work for the project to be considered complete.

#	Feature	Why it is essential
1	Diabetes Q&A Assistant	Allows users to ask diabetes-related questions and receive simple educational answers.
2	Health Information Input	Collects basic user information needed for educational risk-factor assessment.
3	Risk Factor Assessment	Explains possible diabetes risk factors based on the information provided by the user.
4	Data Visualization	Displays health-related data through charts to make information easier to understand.
5	Results Display	Presents answers, risk-factor explanations, and charts in a clear and understandable format.

Out of scope — deliberately excluded. Naming what you will not build is as important as naming what you will; it is how an examiner judges whether your scope is realistic.

The project will not provide medical diagnoses, prescribe medicines, or recommend personalized treatments. It will not connect to hospitals, doctors, or electronic medical records. Real-time monitoring through wearable devices and mobile applications will not be included. The system will focus only on diabetes awareness, educational risk-factor assessment, and basic health data visualization. Advanced clinical prediction and professional medical decision-making are also outside the scope of this project.

5. Data Source

Identify exactly where your data will come from, and confirm that you can actually obtain it. A project whose data source turns out to be unreachable in week 12 cannot be rescued.

Source name and URL	1. Early Stage Diabetes Risk Prediction Dataset – UCI Machine Learning Repository: https://archive.ics.uci.edu/dataset/529/early+stage+diabetes+risk+prediction+dataset 2. CDC Diabetes Information: https://www.cdc.gov/diabetes/about/
Type	Downloadable dataset , Self-collected / generated (for Q and A)
Access requirements	No key needed
Approximate size	The UCI dataset contains 520 records and 16 input features, with a small downloadable file of approximately 34 KB. The educational Q&A collection will be stored locally in a simple text or CSV file.
Licence / terms of use	The UCI dataset is available under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence. The dataset source will be credited in the project. CDC educational information will be used with appropriate source attribution.
Offline fallback	The downloaded UCI dataset will be saved locally in CSV format, and the educational Q&A collection will be stored in a local file. The program will read these files directly from the computer, allowing the risk-factor assessment, charts, and predefined educational answers to work without an internet connection.

> ***Required:*** your program must still run when the network is unavailable, by reading a locally cached copy of the data. Describe your caching plan in the last row above.

6. Technologies and Lab Coverage

For each row, state in one line how your project will use that concept. If you genuinely cannot see a use for one, write "to be identified" — but every row must be satisfied by the time the report is written.

Lab	Concept to be applied	How your project will use it
01–03	Core Python, control flow, data structures	Python variables, conditions, loops, lists, and dictionaries will manage user inputs and diabetes-related information.
04	Functions, lambda, map / filter	Functions will process user information, while filtering will help select relevant records and risk factors.
05	Classes, inheritance, encapsulation, polymorphism	Classes will organize the health assistant, risk assessment, and visualization components into reusable modules.
06	Files (TXT / CSV / JSON), exceptions, web API	The project will read local CSV datasets and text or JSON files, with exception handling for missing or invalid data. A live web API will not be required.
07	NumPy arrays, broadcasting, vectorisation	NumPy will perform efficient numerical operations on health-related data.
08	Pandas — loading, cleaning, grouping	Pandas will load the diabetes dataset, handle missing or duplicate records where applicable, and group data for analysis.
09	Matplotlib / Seaborn visualisations	Matplotlib and Seaborn will create charts showing patterns and distributions in the diabetes dataset.
10–11	Machine learning, model comparison	Suitable classification models will be trained and compared on the diabetes dataset for educational risk-pattern analysis.
12 or 13	Chosen track — NLP text features or image preprocessing	NLP techniques will process users' text questions and help the health assistant identify relevant diabetes-related information.

Lab	Concept to be applied	How your project will use it
14	Documentation, modularity, ethical considerations	The project will use organized code, clear documentation, appropriate source attribution, and a disclaimer that it does not provide medical diagnoses.

7. Division of Work

Member	Modules / components owned	Approximate share
Individual Project Developer HARAM WAQAR	Health Assistant, Risk Assessment, Data Visualization, Dataset Loading and Cleaning, NLP Implementation, Machine Learning Model Comparison, Testing, and Documentation	100%

8. Milestones and Timeline

#	Milestone — what will be demonstrably working	Target date	Owner
1	Dataset downloaded, loaded, and cleaned	29 Oct 2026	Haram
2	NLP Health Assistant answers basic questions	02 Nov 2026	Haram
3	Risk Assessment module works with user input	06 Nov 2026	Haram
4	Charts and machine learning model comparison completed	10 Nov 2026	Haram
5	Final integration, documentation and repository cleanup	14 Nov 2026	Haram

9. Expected Outcome and Risks

The final submission will be a working Python-based GlucoGuide AI system in Jupyter Notebook. The examiner will be able to ask diabetes-related questions, enter basic health information for educational risk-factor assessment, and view charts generated from the dataset. The project will demonstrate Python programming, NLP, data analysis, visualization, and basic machine learning model comparison.

Risk 1: Dataset or data quality issues. If the dataset contains missing values or causes processing errors, I will clean the data and handle errors before analysis.

Risk 2: Limited NLP accuracy. If the assistant cannot answer some questions correctly, I will improve its locally stored question-and-answer collection and provide clear fallback responses.

For office use — Lab Instructor	
Proposal status	Approved ☐ Approved with changes ☐ Revision required ☐
Comments	
Signature and date	

End of Project Proposal